

**Notes: Foster, Haltiwanger, and Syverson (AER, 2008)**  
**Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability?**

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# 1 Big picture

- **Question.** Does market selection operate on physical productivity, revenue productivity, or profitability? The paper asks whether earlier work using revenue-based productivity confounds technical efficiency with plant-level prices and demand.
- **Setting.** The authors study US manufacturing plants producing eleven relatively homogeneous products, including boxes, bread, carbon black, coffee, concrete, hardwood flooring, gasoline, block ice, processed ice, plywood, and raw cane sugar.
- **Core measurement advantage.** Unlike most plant-level productivity studies, the paper observes both physical output and revenue. This allows the authors to construct quantity-based productivity, revenue productivity, prices, and plant-specific demand shocks.
- **Central empirical distinction.** Revenue productivity equals physical productivity plus price in logs. Therefore high revenue productivity can reflect high technical efficiency, high prices, strong demand, or some combination of these forces.
- **Main finding 1: price-productivity relation.** Physical productivity is strongly negatively correlated with plant prices, while revenue productivity is positively correlated with prices. This pattern is consistent with efficient producers charging lower prices because of lower marginal costs.
- **Main finding 2: entry.** Entrants and young plants look less impressive when using revenue productivity because they charge lower prices. In physical productivity terms, entrants have a larger advantage than standard revenue-based measures suggest.
- **Main finding 3: exit and selection.** Low revenue productivity predicts exit, but when revenue productivity is decomposed, both physical productivity and demand matter. Demand shocks are especially powerful predictors of survival.
- **Aggregate implication.** Revenue-based productivity growth decompositions understate the contribution of entry to aggregate productivity because entrants' lower prices obscure their physical productivity advantage.
- **Economic takeaway.** Selection is more accurately understood as selection on profitability, which is jointly determined by technical efficiency and demand, rather than selection on physical productivity alone.

# 2 Incremental contribution of the paper

- **From revenue productivity to physical productivity.** Earlier establishment-level work usually deflated revenue with industry prices and called the result productivity. This paper directly separates physical output, plant prices, and revenue, showing what is hidden inside revenue-based productivity.
- **Direct evidence on price dispersion.** The paper shows that within narrowly defined homogeneous product markets, price dispersion is economically meaningful and systematically related to productivity and demand.

- **A sharper interpretation of market selection.** The paper changes the interpretation of the productivity-survival link: exit is not simply about technical inefficiency; it is also about weak plant-specific demand.
- **Demand estimation contribution.** By using physical productivity as an instrument for price, the paper estimates product-specific demand curves and constructs plant-level demand shocks.
- **Entry and aggregate productivity contribution.** The paper explains why standard revenue-based methods understate entrants' productivity contribution: new and young producers tend to charge lower prices.
- **Connection to misallocation and reallocation.** The findings imply that analyses of reallocation, turnover, and aggregate productivity can be distorted when revenue productivity is interpreted as pure technical efficiency.
- **Methodological benchmark.** The paper becomes a key reference for later work distinguishing TFPR and TFPQ, including studies of markups, demand shocks, organization, and trade-induced reallocation.

### 3 Data and measurement

#### 3.1 Data source and sample

- The paper uses the US Census of Manufactures for census years 1977, 1982, 1987, 1992, and 1997.
- The unit of observation is the establishment, or plant.
- The sample contains 17,669 plant-year observations after applying sample restrictions.
- The authors focus on eleven products with relatively homogeneous physical output, so that output quantities are meaningfully comparable across producers.
- Plants must obtain at least 50 percent of revenue from the focal product, reducing measurement problems from multi-product production.

#### 3.2 Output, price, and productivity measures

For plant  $i$  in year  $t$ , the authors construct a TFP index of the form

$$tfp_{it} = y_{it} - \alpha_l l_{it} - \alpha_k k_{it} - \alpha_m m_{it} - \alpha_e e_{it},$$

where  $y$  is output,  $l$  labor hours,  $k$  capital,  $m$  materials, and  $e$  energy. Factor elasticities are measured using average cost shares.

- **Physical productivity (TFPQ).** Uses physical quantity as output. This is the closest measure to technical efficiency.
- **Revenue productivity (TFPR).** Uses product-level nominal revenue deflated by the sample product price index. In logs,  $TFPR = TFPQ + price$ .

- **Traditional productivity (TFPT).** Uses the conventional revenue-based output measure deflated by an industry-level deflator.
- **Price.** Unit price equals plant revenue divided by plant physical quantity, adjusted to a common 1987 basis.

### 3.3 Demand shock construction

The paper estimates product-specific demand equations of the form

$$\ln q_{it} = \alpha_0 + \alpha_1 \ln p_{it} + \sum_t \alpha_t Year_t + \alpha_2 \ln Income_{mt} + \eta_{it}.$$

The residual component, plus the local income component, is interpreted as a plant-specific demand shock. The key empirical challenge is simultaneity between price and demand. The authors use physical productivity as an instrument for price because more efficient plants should have lower marginal costs and therefore lower prices, but TFPQ should not directly reflect contemporaneous demand once controls are included.

## 4 Model and economic intuition

### 4.1 Selection on profitability

The theoretical model has heterogeneous producers that differ in physical productivity, input costs, and idiosyncratic demand. Profitability depends on all of these dimensions. The model therefore clarifies why selection on profits is not identical to selection on physical productivity.

### 4.2 Price as a mixed signal

A plant with a high price may have strong idiosyncratic demand, high market power, high input costs, or low productivity. Therefore price is not simply a measurement nuisance: it is an economically meaningful object that reflects both demand and supply fundamentals.

### 4.3 Revenue productivity as a confounded measure

Revenue productivity is high when physical productivity is high, but also when prices are high. If efficient plants pass cost advantages through to lower prices, then revenue productivity can understate true technical efficiency. This is especially important for entrants and young plants.

## 5 Empirical specifications

### 5.1 Evolution regressions

The paper compares entrants, exiters, incumbents, and plants of different ages using regressions of the form

$$x_{it} = \beta_0 + \beta_1 Exit_{it} + \beta_2 Entry_{it} + \sum_{PT} \beta_{PT} IndYear_{PT} + u_{it},$$

where  $x_{it}$  is plant-level productivity, price, or demand. Product-year fixed effects ensure that comparisons are within narrowly defined product-year cells.

## 5.2 Exit regressions

To study selection, the authors estimate probit models of plant exit by the next census. The key explanatory variables are TFPT, TFPR, TFPQ, price, demand shock, and capital stock. The main question is whether physical productivity and demand have independent survival effects once revenue productivity is decomposed.

## 5.3 Aggregate productivity decompositions

The paper decomposes industry productivity growth over five-year horizons into within-plant, between-plant, cross, entry, exit, and net-entry components. The comparison across TFPT, TFPR, and TFPQ reveals how price variation changes conclusions about the role of entry and reallocation.

# 6 Identification and empirical design

## 6.1 What the design identifies

- The paper does not randomly assign productivity or demand shocks.
- Its core strength is measurement: it directly observes physical quantities and prices in industries where physical output is relatively homogeneous.
- The key empirical interpretation comes from decomposing revenue productivity into physical productivity and prices, and then decomposing price variation into demand and productivity-related components.

## 6.2 Why product homogeneity matters

If plant prices mainly reflected vertical quality differences, physical output comparisons would be misleading. The authors therefore choose products for which physical units are close substitutes, making TFPQ a more credible measure of technical efficiency.

## 6.3 Main limitations

- The sample is limited to selected homogeneous manufacturing products, so the results may not generalize mechanically to differentiated or service industries.
- Input prices are not fully observed, so some variation in TFPQ may reflect input-cost variation rather than pure technology.
- Demand shocks are estimated residuals, so their interpretation depends on the demand specification and instrument validity.

## 7 Tables and figures

### 7.1 Table 1: Summary statistics for output, price, and productivity measures

Read:

- The table reports correlations and standard deviations after removing product-year fixed effects.
- Traditional, revenue, and physical output are almost perfectly correlated because plant scale is highly persistent across output concepts.
- Physical productivity has a standard deviation of 0.26, larger than revenue productivity's 0.22.
- Price is strongly negatively correlated with physical productivity, with correlation  $-0.54$ .
- Revenue productivity is positively correlated with price, with correlation 0.16.

**Economic message:** The table gives the central measurement insight. Revenue productivity is not the same as technical efficiency because the price component moves in the opposite direction from physical productivity.

TABLE 1—SUMMARY STATISTICS FOR OUTPUT, PRICE, AND PRODUCTIVITY MEASURES

| Correlations        |                   |                   |                    |       |                |                |                 |         |
|---------------------|-------------------|-------------------|--------------------|-------|----------------|----------------|-----------------|---------|
| Variables           | Trad'l.<br>output | Revenue<br>output | Physical<br>output | Price | Trad'l.<br>TFP | Revenue<br>TFP | Physical<br>TFP | Capital |
| Traditional output  | 1.00              |                   |                    |       |                |                |                 |         |
| Revenue output      | 0.99              | 1.00              |                    |       |                |                |                 |         |
| Physical output     | 0.98              | 0.99              | 1.00               |       |                |                |                 |         |
| Price               | -0.03             | -0.03             | -0.19              | 1.00  |                |                |                 |         |
| Traditional TFP     | 0.19              | 0.18              | 0.15               | 0.13  | 1.00           |                |                 |         |
| Revenue TFP         | 0.17              | 0.21              | 0.18               | 0.16  | 0.86           | 1.00           |                 |         |
| Physical TFP        | 0.17              | 0.20              | 0.28               | -0.54 | 0.64           | 0.75           | 1.00            |         |
| Capital             | 0.86              | 0.85              | 0.84               | -0.04 | 0.00           | -0.00          | 0.03            | 1.00    |
| Standard deviations |                   |                   |                    |       |                |                |                 |         |
|                     | 1.03              | 1.03              | 1.05               | 0.18  | 0.21           | 0.22           | 0.26            | 1.14    |

*Notes:* This table shows correlations and standard deviations for plant-level variables for our pooled sample of 17,669 plant-year observations. We remove product-year fixed effects from each variable before computing the statistics. All

Table 1: Summary statistics for output, price, and productivity measures.

### 7.2 Table 2: Estimating price elasticities by product

Read:

- The table estimates demand curves separately by product.
- The IV specification uses physical productivity as an instrument for price.
- Most IV price elasticities are negative and below  $-1$ , consistent with price-setting producers facing elastic demand.
- Concrete has a very elastic estimate, while carbon black is an exception with a small and imprecise IV elasticity.
- The table is the basis for constructing plant-specific demand shocks used later in the selection analysis.

**Economic message:** The demand estimates let the authors separate price movements due to physical efficiency from price movements due to idiosyncratic demand.

TABLE 2—ESTIMATING PRICE ELASTICITIES BY PRODUCT

| Product           | IV estimation                    |                                   | OLS estimation                   |                                   |
|-------------------|----------------------------------|-----------------------------------|----------------------------------|-----------------------------------|
|                   | Price coefficient ( $\alpha_1$ ) | Income coefficient ( $\alpha_2$ ) | Price coefficient ( $\alpha_1$ ) | Income coefficient ( $\alpha_2$ ) |
| Boxes             | -3.02<br>(0.17) [0.61]           | -0.03<br>(0.02)                   | -2.19<br>(0.12)                  | -0.03<br>(0.02)                   |
| Bread             | -3.09<br>(0.42) [0.33]           | 0.12<br>(0.05)                    | -0.89<br>(0.15)                  | 0.07<br>(0.04)                    |
| Carbon black      | -0.52<br>(0.38) [0.50]           | -0.21<br>(0.11)                   | -0.57<br>(0.21)                  | -0.21<br>(0.11)                   |
| Coffee            | -3.63<br>(0.98) [0.41]           | 0.22<br>(0.14)                    | -1.03<br>(0.32)                  | 0.20<br>(0.13)                    |
| Concrete          | -5.93<br>(0.36) [0.10]           | 0.13<br>(0.01)                    | -0.83<br>(0.09)                  | 0.15<br>(0.01)                    |
| Hardwood flooring | -1.67<br>(0.48) [0.61]           | -0.20<br>(0.18)                   | -0.87<br>(0.47)                  | -0.24<br>(0.18)                   |
| Gasoline          | -1.42<br>(2.72) [0.20]           | 0.23<br>(0.07)                    | -0.16<br>(0.80)                  | 0.23<br>(0.07)                    |
| Block ice         | -2.05<br>(0.46) [0.32]           | 0.00<br>(0.11)                    | -0.63<br>(0.20)                  | 0.16<br>(0.07)                    |
| Processed ice     | -1.48<br>(0.27) [0.37]           | 0.18<br>(0.03)                    | -0.70<br>(0.13)                  | 0.16<br>(0.03)                    |
| Plywood           | -1.21<br>(0.14) [0.89]           | -0.23<br>(0.10)                   | -1.19<br>(0.13)                  | -0.23<br>(0.10)                   |
| Sugar             | -2.52<br>(1.01) [0.15]           | 0.76<br>(0.13)                    | -1.04<br>(0.55)                  | 0.72<br>(0.12)                    |

Notes: This table shows the results of estimating demand isoelastic curves separately for each product (shown by row). Two specifications are estimated for each product: one using IV methods and one using OLS for comparison.

Table 2: Estimating price elasticities by product.

### 7.3 Table 3: Persistence of productivity, prices, and demand shocks

Read:

- Productivity, prices, and demand shocks are persistent at the plant level.
- Persistence supports the idea that plant heterogeneity is not purely transitory noise.
- The persistence of demand matters because survival depends not only on current demand but also on whether demand advantages are expected to persist.

**Economic message:** Selection can operate on persistent plant-level demand and productivity types, not just short-run shocks.

TABLE 3—PERSISTENCE OF PRODUCTIVITY, PRICES AND DEMAND SHOCKS

| Dependent variable | Five-year horizon     |                     | Implied one-year persistence rates |                     |
|--------------------|-----------------------|---------------------|------------------------------------|---------------------|
|                    | Unweighted regression | Weighted regression | Unweighted regression              | Weighted regression |
| Traditional TFP    | 0.249<br>(0.017)      | 0.316<br>(0.042)    | 0.757                              | 0.794               |
| Revenue TFP        | 0.277<br>(0.021)      | 0.316<br>(0.042)    | 0.774                              | 0.794               |
| Physical TFP       | 0.312<br>(0.019)      | 0.358<br>(0.049)    | 0.792                              | 0.814               |
| Price              | 0.365<br>(0.025)      | 0.384<br>(0.066)    | 0.817                              | 0.826               |
| Demand shock       | 0.619<br>(0.013)      | 0.843<br>(0.021)    | 0.909                              | 0.966               |

Notes: This table reports the results of regressing a plant's current TFP, price, or idiosyncratic demand (shown by row) on its value in the previous Census of Manufactures (five years prior). Reported coefficients are those on the lagged variable. The sample includes continuing establishments only,  $N = 7812$ . Weighted regressions are weighted by revenue. The implied one-year persistence rates are the autocorrelation coefficients for annual data that would yield the same persistence over a five-year period as is implied by the regressions; i.e., they are the estimated coefficients to the one-fifth power. Standard errors, clustered by plant, are in parentheses.

Table 3: Persistence of productivity, prices, and demand shocks.

## 7.4 Tables 4 and 5: Entry, exit, and age effects

Read Table 4:

- Exiting plants have lower productivity and much lower demand shocks than continuing plants.
- Entrants have higher physical productivity in weighted regressions but lower prices and lower demand shocks.
- Entrants' revenue productivity advantage is much smaller than their physical productivity advantage.

Read Table 5:

- Young and medium-aged plants charge lower prices than old incumbents.
- In weighted regressions, entrants have substantially higher physical productivity but lower revenue productivity.
- The age pattern shows that the revenue-productivity understatement is not only an entry-year phenomenon; it also applies to young producers more generally.

**Economic message:** New and young plants are physically efficient but price aggressively. Revenue productivity therefore understates their technical advantage.

TABLE 4—EVOLUTION OF REVENUE PRODUCTIVITY, PHYSICAL PRODUCTIVITY, PRICES AND DEMAND SHOCKS

| Variable        | Unweighted regression |                     | Weighted regression |                     |
|-----------------|-----------------------|---------------------|---------------------|---------------------|
|                 | Exit dummy            | Entry dummy         | Exit dummy          | Entry dummy         |
| Traditional TFP | -0.0209<br>(0.0042)   | 0.0014<br>(0.0040)  | -0.0164<br>(0.0126) | -0.0032<br>(0.0188) |
| Revenue TFP     | -0.0218<br>(0.0044)   | 0.0110<br>(0.0042)  | -0.0197<br>(0.0135) | -0.0005<br>(0.0183) |
| Physical TFP    | -0.0186<br>(0.0050)   | 0.0125<br>(0.0047)  | -0.0142<br>(0.0144) | 0.0383<br>(0.0177)  |
| Price           | -0.0033<br>(0.0031)   | -0.0015<br>(0.0028) | -0.0055<br>(0.0080) | -0.0388<br>(0.0141) |
| Demand shock    | -0.3586<br>(0.0228)   | -0.3976<br>(0.0224) | -0.5903<br>(0.0968) | -0.2188<br>(0.1278) |

Notes: This table shows the coefficients on indicator variables for exiting and entering plants (shown by column) when we regress plant-level productivity, price, and demand values (shown by row) on these indicators, and a full set of product-year fixed effects. The sample is our pooled sample of 17,314 plant-year observations (355 observations from the main sample are excluded because we cannot determine exiting plants in the 1997 CM, the final year of observa-

(a) Table 4: Entry and exit.

TABLE 5—EVOLUTION OF PRODUCTIVITY, PRICE AND DEMAND WITH AGE EFFECTS

| Variable               | Plant age dummies   |                     |                     |                     |
|------------------------|---------------------|---------------------|---------------------|---------------------|
|                        | Exit                | Entry               | Young               | Medium              |
| Unweighted regressions |                     |                     |                     |                     |
| Traditional TFP        | -0.0211<br>(0.0042) | 0.0044<br>(0.0044)  | 0.0074<br>(0.0048)  | 0.0061<br>(0.0048)  |
| Revenue TFP            | -0.0220<br>(0.0044) | 0.0133<br>(0.0047)  | 0.0075<br>(0.0051)  | 0.0028<br>(0.0053)  |
| Physical TFP           | -0.0186<br>(0.0050) | 0.0128<br>(0.0053)  | 0.0046<br>(0.0058)  | -0.0039<br>(0.0062) |
| Price                  | -0.0034<br>(0.0031) | 0.0005<br>(0.0034)  | 0.0029<br>(0.0038)  | 0.0067<br>(0.0042)  |
| Demand shock           | -0.3466<br>(0.0227) | -0.5557<br>(0.0264) | -0.3985<br>(0.0263) | -0.3183<br>(0.0267) |
| Weighted regressions   |                     |                     |                     |                     |
| Traditional TFP        | -0.0156<br>(0.0127) | -0.0068<br>(0.0203) | -0.0156<br>(0.0171) | -0.0234<br>(0.0132) |
| Revenue TFP            | -0.0191<br>(0.0136) | -0.0038<br>(0.0200) | -0.0180<br>(0.0198) | -0.0165<br>(0.0131) |
| Physical TFP           | -0.0142<br>(0.0144) | 0.0383<br>(0.0186)  | 0.0056<br>(0.0142)  | -0.0050<br>(0.0135) |
| Price                  | -0.0049<br>(0.0079) | -0.0421<br>(0.0147) | -0.0236<br>(0.0114) | -0.0114<br>(0.0096) |
| Demand shock           | -0.5790<br>(0.0972) | -0.2785<br>(0.1459) | -0.3133<br>(0.1695) | -0.3164<br>(0.1197) |

Notes: This table shows the coefficients on indicator variables for exiting, entering, and continuing plants of two age cohorts (shown by column: "youne" establishments first appeared in the census five years ago, "medium" establish-

(b) Table 5: Age effects.

Tables 4 and 5: Evolution of productivity, prices, and demand.

## 7.5 Table 6: Selection on productivity or profitability?

Read:

- Traditional and revenue productivity both predict lower exit probabilities when used alone.
- When physical productivity and prices are included together, both matter more than when each is considered separately.
- Demand shocks are a very strong survival predictor.

- The demand effect remains strong after controlling for capital stock.

**Economic message:** The table is the paper’s core selection result. Plants exit not simply because they are technically inefficient, but because they combine weak productivity, weak prices, and especially weak demand.

| TABLE 6—SELECTION ON PRODUCTIVITY OR PROFITABILITY? |                   |                   |                   |                   |                   |                   |                   |
|-----------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Specification:                                      | (1)               | (2)               | (3)               | (4)               | (5)               | (6)               | (7)               |
| Traditional TFP                                     | −0.073<br>(0.015) |                   |                   |                   |                   |                   |                   |
| Revenue TFP                                         |                   | −0.063<br>(0.014) |                   |                   |                   |                   |                   |
| Physical TFP                                        |                   |                   | −0.040<br>(0.012) |                   |                   | −0.062<br>(0.014) | −0.034<br>(0.012) |
| Prices                                              |                   |                   |                   | −0.021<br>(0.018) |                   | −0.069<br>(0.021) |                   |
| Demand shock                                        |                   |                   |                   |                   | −0.047<br>(0.003) |                   | −0.047<br>(0.003) |
| Controlling for plant capital stock                 |                   |                   |                   |                   |                   |                   |                   |
| Traditional TFP                                     | −0.069<br>(0.015) |                   |                   |                   |                   |                   |                   |
| Revenue TFP                                         |                   | −0.061<br>(0.013) |                   |                   |                   |                   |                   |
| Physical TFP                                        |                   |                   | −0.035<br>(0.012) |                   |                   | −0.059<br>(0.014) | −0.034<br>(0.012) |
| Prices                                              |                   |                   |                   | −0.030<br>(0.018) |                   | −0.076<br>(0.021) |                   |
| Demand shock                                        |                   |                   |                   |                   | −0.030<br>(0.004) |                   | −0.029<br>(0.004) |
| Capital stock                                       | −0.046<br>(0.003) | −0.046<br>(0.003) | −0.046<br>(0.003) | −0.046<br>(0.003) | −0.023<br>(0.004) | −0.046<br>(0.003) | −0.023<br>(0.004) |

Table 6: Selection on productivity or profitability?

## 7.6 Table 7: Decomposition of industry productivity growth

**Read:**

- Aggregate revenue and physical productivity growth are mechanically equal for TFPR and TFPQ under the product-level deflator, but their components differ.
- Entry contributes more to physical productivity growth than to revenue productivity growth.
- The net-entry contribution is larger for physical productivity because entrants have lower prices that depress their revenue productivity.

**Economic message:** Standard revenue-based decompositions understate the contribution of entry to true productivity growth.

## 7.7 Tables 8 and 9: Robustness checks

**Read Table 8:**

- Restricting the sample to local-market products preserves the core selection findings.
- Demand shocks remain an important determinant of survival.
- The decomposition still shows a larger entry contribution for physical productivity than for revenue productivity.

**Read Table 9:**

TABLE 7—DECOMPOSITION OF INDUSTRY PRODUCTIVITY GROWTH OVER FIVE-YEAR HORIZONS

| Productivity measure | Total growth | Within | Components of decomposition (BHC/FHK) |       |       |      | Net entry |
|----------------------|--------------|--------|---------------------------------------|-------|-------|------|-----------|
|                      |              |        | Between                               | Cross | Entry | Exit |           |
| Traditional          | 2.30         | 0.82   | −0.38                                 | 1.16  | 0.55  | 0.16 | 0.70      |
| Revenue              | 5.13         | 3.34   | −0.52                                 | 1.39  | 0.73  | 0.19 | 0.93      |
| Physical             | 5.13         | 3.44   | −0.41                                 | 0.76  | 1.23  | 0.12 | 1.35      |
| Productivity measure | Total growth | Within | Components of decomposition (GR)      |       |       |      | Net entry |
|                      |              |        | Between                               | Cross | Entry | Exit |           |
| Traditional          | 2.30         | 1.40   | 0.18                                  |       | 0.44  | 0.27 | 0.72      |
| Revenue              | 5.13         | 4.03   | 0.16                                  |       | 0.55  | 0.39 | 0.94      |
| Physical             | 5.13         | 3.82   | −0.05                                 |       | 1.04  | 0.32 | 1.36      |

Notes: This table shows decompositions of industry-level productivity growth for three different productivity measures (shown by row) using equations (12) and (13) in text. The column labeled “Total Growth” reflects the weighted average five-year productivity growth for the industry. The remaining columns reflect the individual terms in the decomposition. Weights used in decompositions are revenue weights. Average industry revenues across the sample are used to calculate the results for the average industry. See text for details.

Table 7: Decomposition of industry productivity growth.

- Restricting to highly specialized producers produces similar qualitative results.
- Selection coefficients for demand shocks and physical productivity remain strong.
- The robustness checks suggest that multi-product measurement problems do not drive the main findings.

**Economic message:** The central conclusions survive two important sample restrictions: local-market products and highly specialized producers.

TABLE 8—ROBUSTNESS CHECK: LOCAL-MARKET PRODUCTS ONLY

| A. Selection on productivity or profitability?                           |                   |                   |                                       |                   |                   |                   |                   |
|--------------------------------------------------------------------------|-------------------|-------------------|---------------------------------------|-------------------|-------------------|-------------------|-------------------|
| Specification:                                                           | (1)               | (2)               | (3)                                   | (4)               | (5)               | (6)               | (7)               |
| Traditional TFP                                                          | −0.076<br>(0.016) |                   |                                       |                   |                   |                   |                   |
| Revenue TFP                                                              |                   | −0.067<br>(0.014) |                                       |                   |                   |                   |                   |
| Physical TFP                                                             |                   |                   | −0.055<br>(0.013)                     |                   |                   | −0.069<br>(0.014) | −0.048<br>(0.013) |
| Prices                                                                   |                   |                   |                                       | −0.002<br>(0.020) |                   | −0.052<br>(0.023) |                   |
| Demand shock                                                             |                   |                   |                                       |                   | −0.046<br>(0.003) |                   | −0.046<br>(0.003) |
| Controlling for plant capital stock                                      |                   |                   |                                       |                   |                   |                   |                   |
| Traditional TFP                                                          | −0.072<br>(0.016) |                   |                                       |                   |                   |                   |                   |
| Revenue TFP                                                              |                   | −0.064<br>(0.014) |                                       |                   |                   |                   |                   |
| Physical TFP                                                             |                   |                   | −0.049<br>(0.013)                     |                   |                   | −0.065<br>(0.014) | −0.048<br>(0.013) |
| Prices                                                                   |                   |                   |                                       | −0.014<br>(0.020) |                   | −0.061<br>(0.023) |                   |
| Demand shock                                                             |                   |                   |                                       |                   | −0.028<br>(0.004) |                   | −0.028<br>(0.004) |
| Capital stock                                                            | −0.045<br>(0.003) | −0.046<br>(0.003) | −0.045<br>(0.003)                     | −0.046<br>(0.003) | −0.024<br>(0.004) | −0.046<br>(0.003) | −0.024<br>(0.004) |
| B. Decomposition of industry productivity growth over five-year horizons |                   |                   |                                       |                   |                   |                   |                   |
| Productivity measure                                                     | Total growth      | Within            | Components of decomposition (BHC/FHK) |                   |                   |                   | Net entry         |
|                                                                          |                   |                   | Between                               | Cross             | Entry             | Exit              |                   |
| Traditional                                                              | 0.84              | −0.43             | −0.34                                 | 0.91              | 0.58              | 0.11              | 0.69              |
| Revenue                                                                  | 4.01              | 2.39              | −0.45                                 | 1.12              | 0.80              | 0.15              | 0.95              |
| Physical                                                                 | 4.01              | 2.37              | −0.33                                 | 0.57              | 1.29              | 0.11              | 1.40              |

(a) Table 8: Local-market products.

TABLE 9—ROBUSTNESS CHECK: HIGHLY SPECIALIZED PRODUCERS ONLY

| A. Selection on productivity or profitability?                           |                   |                   |                                       |                   |                   |                   |                   |
|--------------------------------------------------------------------------|-------------------|-------------------|---------------------------------------|-------------------|-------------------|-------------------|-------------------|
| Specification:                                                           | (1)               | (2)               | (3)                                   | (4)               | (5)               | (6)               | (7)               |
| Traditional TFP                                                          | −0.089<br>(0.018) |                   |                                       |                   |                   |                   |                   |
| Revenue TFP                                                              |                   | −0.071<br>(0.017) |                                       |                   |                   |                   |                   |
| Physical TFP                                                             |                   |                   | −0.048<br>(0.014)                     |                   |                   | −0.071<br>(0.017) | −0.040<br>(0.014) |
| Prices                                                                   |                   |                   |                                       | −0.018<br>(0.021) |                   | −0.072<br>(0.025) |                   |
| Demand shock                                                             |                   |                   |                                       |                   | −0.047<br>(0.003) |                   | −0.047<br>(0.003) |
| Controlling for plant capital stock                                      |                   |                   |                                       |                   |                   |                   |                   |
| Traditional TFP                                                          | −0.085<br>(0.018) |                   |                                       |                   |                   |                   |                   |
| Revenue TFP                                                              |                   | −0.069<br>(0.016) |                                       |                   |                   |                   |                   |
| Physical TFP                                                             |                   |                   | −0.042<br>(0.014)                     |                   |                   | −0.067<br>(0.017) | −0.040<br>(0.014) |
| Prices                                                                   |                   |                   |                                       | −0.029<br>(0.021) |                   | −0.081<br>(0.025) |                   |
| Demand shock                                                             |                   |                   |                                       |                   | −0.031<br>(0.004) | −0.031<br>(0.004) |                   |
| Capital stock                                                            | −0.045<br>(0.003) | −0.045<br>(0.003) | −0.045<br>(0.003)                     | −0.045<br>(0.003) | −0.021<br>(0.004) | −0.045<br>(0.003) | −0.021<br>(0.004) |
| B. Decomposition of industry productivity growth over five-year horizons |                   |                   |                                       |                   |                   |                   |                   |
| Productivity measure                                                     | Total growth      | Within            | Components of decomposition (BHC/FHK) |                   |                   |                   | Net entry         |
|                                                                          |                   |                   | Between                               | Cross             | Entry             | Exit              |                   |
| Traditional                                                              | 5.87              | 3.36              | −0.42                                 | 2.23              | 0.44              | 0.25              | 0.69              |
| Revenue                                                                  | 6.90              | 3.98              | −0.76                                 | 2.77              | 0.66              | 0.22              | 0.87              |
| Physical                                                                 | 6.90              | 4.67              | −0.97                                 | 1.76              | 1.13              | 0.27              | 1.40              |

(b) Table 9: Specialized producers.

Tables 8 and 9: Robustness checks.

## 8 Short summary

- The paper studies plant-level reallocation in industries where physical quantities and plant prices are observed.
- It distinguishes physical productivity (TFPQ), revenue productivity (TFPR), traditional revenue productivity (TFPT), prices, and demand shocks.
- Physical productivity is negatively correlated with plant prices, while revenue productivity is positively correlated with prices.
- Entrants and young plants charge lower prices, so revenue productivity understates their physical productivity advantage.
- Exit is predicted by low revenue productivity, but the decomposition shows that physical productivity and demand have separate effects.
- Demand shocks are especially important for survival.
- Aggregate productivity decompositions based on revenue productivity understate entry's contribution to true productivity growth.
- The paper's main message is that selection is on profitability, not productivity alone.

## 9 Conclusion

### 9.1 Core conclusion

The paper shows that revenue-based productivity can be a misleading proxy for technical efficiency. Because revenue productivity combines physical productivity and price, it confounds supply-side efficiency with demand-side conditions.

### 9.2 Economic intuition

Efficient plants often charge lower prices. Entrants and young plants, in particular, appear to use lower prices while being physically productive. Standard revenue-based productivity measures therefore make entrants look less productive than they really are from a technical-efficiency perspective.

### 9.3 Incremental contribution revisited

The paper's key contribution is not only documenting price dispersion. Its deeper contribution is showing how price dispersion changes the interpretation of selection, entry, exit, and aggregate productivity growth. It shifts the empirical question from "do productive plants survive?" to "which components of profitability make plants survive?"

## **9.4 Implications for future research**

The paper suggests that studies using TFPR should be careful when interpreting productivity dispersion, selection, and misallocation. When physical output or plant-level prices are unavailable, revenue productivity may reflect demand, markups, quality, or pricing strategies as much as technical efficiency.

## **9.5 Limitations**

The paper focuses on homogeneous manufacturing products, so its measurement advantage comes with external-validity limits. The authors also do not fully observe input prices, and the demand shocks are estimated rather than directly observed.

## **9.6 Takeaway**

The clean takeaway is that market selection is multi-dimensional: efficient producers survive, but so do producers with strong idiosyncratic demand. Revenue productivity alone masks this distinction.